

1. Derive a formula for the efficiency of the Diesel cycle, in terms of the compression ratio V_1/V_2 and the cutoff ratio V_3/V_2 . Show that for a given compression ratio, the Diesel cycle is less efficient than the Otto cycle. Evaluate the theoretical efficiency of a Diesel engine with a compression ratio of 18 and a cutoff ratio of 2.

We know that $Q = 0$ for both adiabatic processes, for isobaric processes $Q = nC_P\Delta T$, and that for isometric processes $Q = nC_V\Delta T$. We also know that from point 2 to point 3 on the PV diagram for a diesel process, an isobaric process occurs, and that from point 4 to point 1, an isometric process occurs.

$$\begin{aligned}
 \text{Let } c &= \frac{V_1}{V_2} \text{ and } u = \frac{V_3}{V_2} \\
 e_D &= 1 - \frac{Q_{out}}{Q_{in}} \\
 &= 1 - \frac{nC_V(T_4 - T_1)}{nC_P(T_3 - T_2)} \\
 \frac{nC_V}{nC_P} &= \frac{1}{\gamma} \\
 1 \rightarrow 2 : \quad T_1 V_1^{\gamma-1} &= T_2 V_2^{\gamma-1} \\
 T_2 &= \left(\frac{V_1}{V_2}\right)^{\gamma-1} T_1 \\
 &= c^{\gamma-1} T_1 \\
 3 \rightarrow 4 : \quad T_3 V_3^{\gamma-1} &= T_4 V_1^{\gamma-1} \\
 T_4 &= \left(\frac{V_3}{V_1}\right)^{\gamma-1} T_3 \\
 \frac{V_3}{T_3} &= \frac{V_2}{T_2} \\
 T_3 &= T_2 \frac{V_3}{V_2} \\
 &= u T_2 \\
 &= u c^{\gamma-1} T_1 \\
 T_4 &= \left(\frac{V_3}{V_1}\right)^{\gamma-1} u c^{\gamma-1} T_1 \\
 &= \left(\frac{V_3}{V_1}\right)^{\gamma-1} \left(\frac{V_1}{V_2}\right)^{\gamma-1} u T_1 \\
 &= \left(\frac{V_3}{V_2}\right)^{\gamma-1} u T_1 \\
 &= u^\gamma T_1
 \end{aligned}$$

$$e_D = 1 - \frac{1}{\gamma} \frac{u^\gamma T_1 - T_1}{uc^{\gamma-1}T_1 - c^{\gamma-1}T_1}.$$

$$e_D = 1 - \frac{1}{\gamma} \frac{u^\gamma - 1}{c^{\gamma-1}(u-1)}$$

We see that the efficiency for an Otto cycle is

$$e_O = 1 - \frac{1}{c^{\gamma-1}}.$$

We seek to show that $e_D < e_O$ for an arbitrary c . So, we will demonstrate that $e_O - e_D > 0$.

$$\begin{aligned} e_O - e_D &> 0 \\ 1 - \frac{1}{c^{\gamma-1}} - \left(1 - \frac{1}{\gamma} \frac{u^\gamma - 1}{c^{\gamma-1}(u-1)}\right) &> 0 \\ \frac{1}{\gamma} \frac{u^\gamma - 1}{c^{\gamma-1}(u-1)} - \frac{1}{c^{\gamma-1}} &> 0 \\ \frac{1}{c^{\gamma-1}} \left(\frac{1}{\gamma} \frac{u^\gamma - 1}{u-1} - 1\right) &> 0 \end{aligned}$$

We know that for all $V_3 > V_2$, $u > 1$.

Also, we know that $\gamma > 1$.

Now, we seek to show that the quantity $\frac{1}{\gamma} \frac{u^\gamma - 1}{u-1} - 1 > 0$.

$$u^\gamma - 1 > \gamma(u-1)$$

$$u^\gamma - 1 - \gamma(u-1) > 0.$$

If we define $f(u) = u^\gamma - 1 - \gamma(u-1)$, we just need to show that $f(u) > 0$ for $u > 1$.

$$f(1) = 1 - 1 - 0 = 0$$

$$\begin{aligned} f'(u) &= \gamma u^{\gamma-1} - \gamma \\ &= \gamma(u^{\gamma-1} - 1) \end{aligned}$$

We see that $f(u)$ starts at 0 and that its derivative is greater than zero for $u > 1$, so

$$\frac{1}{\gamma} \frac{u^\gamma - 1}{u-1} - 1 > 0.$$

Thus, we conclude that the quantity $\frac{1}{\gamma} \frac{u^\gamma - 1}{u-1} - 1 > 0$, equivalent to $e_O - e_D > 0$.

We see conclusively that the Diesel cycle is less efficient than the Otto cycle.

For $c = 18$ and $u = 2$, we see that:

$$\begin{aligned} e_D &= 1 - \frac{1}{\gamma} \frac{2^\gamma - 1}{18^{\gamma-1}(2-1)} \\ &= 1 - \frac{1}{\gamma} \frac{2^\gamma - 1}{18^{\gamma-1}} \end{aligned}$$

Assuming $\gamma = 1.4$ for air,

$$e_D = 1 - \frac{1}{1.4} \frac{2^{1.4} - 1}{18^{1.4-1}}$$

$$\begin{aligned}
&= 1 - 0.368 \\
&= 0.632
\end{aligned}$$

So, we see that the Diesel engine's theoretical efficiency for $c = 18$ and $u = 2$ is 63%.